Constrained Nonlinear Programming Problem

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1. Introduction

The Lagrangian method converts a constrained optimization problem into an unconstrained one by incorporating equality constraints using multipliers. Solutions are obtained by solving the stationarity and feasibility conditions, followed by second-order tests.

2. Lagrangian Method for Solving NLPP with Equality Constraints

Consider the nonlinear programming problem (NLPP)

$$\begin{aligned} &\min f(x), \\ &\text{subject to } g_i(x) = 0, \qquad i = 1, 2, \ldots, m, \end{aligned}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ and $g_i : \mathbb{R}^n \to \mathbb{R}$ are continuously differentiable functions.

Lagrangian Function

To handle the equality constraints, we introduce Lagrange multipliers

$$\lambda_1, \lambda_2, \ldots, \lambda_m,$$

and define the Lagrangian function:

$$L(x, \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x).$$

First–Order Necessary (KKT) Conditions

A point x^* is a **candidate for optimality** if there exist multipliers λ^* such that:

1. **Stationarity:**
$$\nabla_x L(x^*, \lambda^*) = \nabla f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*) = 0.$$
2. **Primal feasibility:**
$$g_i(x^*) = 0, \qquad i = 1, \ldots, m.$$

These are known as the **Lagrange necessary conditions** (special case of KKT conditions for equality constraints).

Geometric Interpretation

The constraint $g_i(x) = 0$ defines a surface in $\mathbb{R}^n$. At the optimum, the gradient of the objective function $\nabla f(x^*)$ must lie in the span of the constraint gradients:

$$\nabla f(x^*) = -\sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*).$$

This means that $\nabla f(x^*)$ is **orthogonal to all feasible directions**. Thus, the objective cannot be further reduced without violating the constraints.

3. Algorithm: Extremality Test Using Lagrangian and Bordered Hessian Matrix

Consider the nonlinear programming problem:

$$\min / \max \; f(x_1, \dots, x_n) \qquad \text{subject to} \qquad g_i(x) = 0, \; i = 1, \dots, m.$$

Step 1: Form the Lagrangian

Define the Lagrangian function:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x).$$

Step 2: First-Order (KKT) Conditions

Solve the system:

$$\nabla_x \mathcal{L}(x, \lambda) = 0, \qquad g_i(x) = 0.$$

The solutions give candidate stationary points (x^*, λ^*).

Step 3: Compute the Hessian of the Lagrangian

Find the Hessian with respect to x:

$$H = \nabla^2_{xx} \mathcal{L}(x^*, \lambda^*).$$

Step 4: Construct the Bordered Hessian Matrix

Let $\nabla g(x^*)$ denote the Jacobian matrix of the constraints. The bordered Hessian is:

$$H_B = \begin{bmatrix} 0 & \nabla g(x^*)^\top \\ \nabla g(x^*) & H \end{bmatrix}.$$

Step 5: Compute Leading Principal Minors

Let D_k denote the determinant of the top-left $k \times k$ principal minor of H_B. For classification, only the minors starting from index $2m + 1$ are used.

Step 6: Extremality Condition

(a) Local Maximum: The stationary point x^* is a **local maximum** if the principal minor of order $2m+1$ have the sign $(-1)^{m+1}$ and then the remaining minors are in alternating sign.

(b) Local Minimum: The point x^* is a **local minimum** if the sign of all principle minors of order greater than or equal to $2m+1$ is $(-1)^m$.

(c) Saddle Point: If the sign pattern does not match the above cases, then:

$$x^* \text{ is a saddle point.}$$

4. Example

Problem 1 *Solve the following NLPP Minimize*

$$f(x_1, x_2) = x_1^2 + 2x_2^2$$

subject to the equality constraint

$$g(x_1, x_2) = x_1^1 + x_2^2 = 1.$$

Answer: The Lagrangian is given by:

$$L(x_1, x_2, \lambda) = x_1^2 + 2x_2^2 + \lambda(x_1^2 + x_2^2 - 1).$$

First-order conditions:

$$\nabla L = 0 \implies 2x_1(1+\lambda) = 0, \qquad 2x_2(2+\lambda) = 0, \qquad \text{and} \qquad \partial_\lambda L = 0 \implies x_1^1 + x_2^2 - 1 = 0.$$

Solving,

$$\begin{aligned} &\text{if} \quad x_1 = 0 \implies x_2 = \pm 1, \\ &\text{if} \quad \lambda = -1 \implies x_2 = 0, \quad x_1 = \pm 1. \end{aligned}$$

we get

$$\text{if} \quad x_1^* = x_2^* = \frac{1}{2}, \qquad \lambda^* = -1.$$

Thus, the stationary points are

$$(1,0)\,,(-1,0)\,,(0,1)\,,(0,-1)$$

Since,

$$f\,(1,0) = 1 = f\,(-1,0)\,, \text{and } f\,(0,1) = 2 = f\,(0,-1)$$

$(1,0), (-1,0)$ are local minimizer and $(0,1), (0,-1)$ are the local maximizer.

Remark: Since the constarint of the above problem is a close and bounded set, we can check from the functional values of the stationary points. Otherwise, we need to check the extremality from the Boredered Hessian matrix.

Problem 2 *Minimize*

$$f(x_1, x_2) = x_1^2 + x_2^2$$

subject to the equality constraint

$$g(x_1, x_2) = x_1 + x_2 - 1 = 0.$$

Answer:

We want to minimize

$$f(x_1, x_2) = x_1^2 + x_2^2$$

subject to the equality constraint

$$g(x_1, x_2) = x_1 + x_2 - 1 = 0.$$

The Lagrangian function is

$$\mathcal{L}(x_1, x_2, \lambda) = x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1).$$

First–order necessary conditions:

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 + \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 + \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = x_1 + x_2 - 1 = 0.$$

Subtracting the first two equations gives

$$x_1 = x_2.$$

Using the constraint,

$$x_1 + x_1 = 1 \quad \Rightarrow \quad x_1 = x_2 = \frac{1}{2}.$$

Substituting in $2x_1 + \lambda = 0$ gives

$$\lambda = -1.$$

$$\boxed{x_1^* = \tfrac{1}{2}, \quad x_2^* = \tfrac{1}{2}, \quad \lambda^* = -1}$$

The Hessian of the objective function is

$$H_f = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The gradient of the constraint is

$$\nabla g = (1 \;\; 1).$$

Thus the bordered Hessian is

$$H_B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Now we consider the minor of order $2m + 1 = 3$.

$$\Delta_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = -6,$$

and the condition for a minimum with one constraint is

$$\Delta_{2m+1} = -6 < 0.$$

Since the sign of Δ_{2m+1} is $(-1)^m$, the stationary point is a strict local minimum.

$$x_1^* = \frac{1}{2}, \qquad x_2^* = \frac{1}{2}, \qquad f(x_1^*, x_2^*) = \frac{1}{2}.$$

Thus, the point $\left(\frac{1}{2}, \frac{1}{2}\right)$ is a **local minimum**.

Problem 3 *Minimize*

$$f(x_1, x_2) = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$$

subject to the equality constraint

$$x_1 + x_2 + x_3 = 15,$$
$$2x_1 - x_2 + 2x_3 = 20.$$

Answer:
We want to minimize

$$f(x_1, x_2, x_3) = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$$

subject to two equality constraints

$$g_1(x_1, x_2, x_3) = x_1 + x_2 + x_3 - 15 = 0, \qquad g_2(x_1, x_2, x_3) = 2x_1 - x_2 + 2x_3 - 20 = 0.$$

Now we solve the problem step by step.

1. Lagrangian Function

$$\mathcal{L}(x_1, x_2, x_3, \lambda_1, \lambda_2) = 4x_1^2+2x_2^2+x_3^2-4x_1x_2+\lambda_1(x_1+x_2+x_3-15)+\lambda_2(2x_1-x_2+2x_3-20).$$

2. First-Order Necessary Conditions

$$\frac{\partial \mathcal{L}}{\partial x_1} = 8x_1 - 4x_2 + \lambda_1 + 2\lambda_2 = 0,$$
$$\frac{\partial \mathcal{L}}{\partial x_2} = 4x_2 - 4x_1 + \lambda_1 - \lambda_2 = 0,$$
$$\frac{\partial \mathcal{L}}{\partial x_3} = 2x_3 + \lambda_1 + 2\lambda_2 = 0.$$

The constraints are

$$x_1 + x_2 + x_3 = 15, \qquad 2x_1 - x_2 + 2x_3 = 20.$$

3. Solving the System
From the first two derivative equations:

$$8x_1 - 4x_2 = -\lambda_1 - 2\lambda_2, \quad -4x_1 + 4x_2 = -\lambda_1 + \lambda_2, \quad 2x_3 = -\lambda_1 - 2\lambda_2$$

Solving these equations we get

$$x_1 = -\frac{2\lambda_1 + \lambda_2}{4}, \quad x_2 = -\frac{3\lambda_1}{4}, \quad x_3 = -\frac{\lambda_1 + 2\lambda_2}{4} \tag{1}$$

Put all these values into the given constarints:

$$7\lambda_1 + 5\lambda_2 = -60, \quad 5\lambda_1 + 10\lambda_2 = -80$$

Solve them, we get

$$\lambda_1^* = -\frac{40}{9}, \qquad \lambda_2^* = -\frac{52}{9}.$$

Using these value in (1)

$$\boxed{x_1^* = \frac{11}{3}, \qquad x_2^* = \frac{10}{3}, \qquad x_3^* = 8.}$$

5. Nature of the Point (Bordered Hessian Test)
Hessian of the objective:

$$H_f = \begin{pmatrix} 8 & -4 & 0 \\ -4 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Gradients of constraints:

$$\nabla g_1 = (1 \ \ 1 \ \ 1), \qquad \nabla g_2 = (2 \ \ -1 \ \ 2).$$

Bordered Hessian:

$$H_B = \begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & -1 & 2 \\ 1 & 2 & 8 & -4 & 0 \\ 1 & -1 & -4 & 4 & 0 \\ 1 & 2 & 0 & 0 & 2 \end{pmatrix}.$$

For minimization with two constraints, so $m = 2$. Now check:

$$\Delta_{2m+1} = \Delta_5 = 90 > 0.$$

The sign is $(-1)^m$.

$$\boxed{\text{The stationary point is a strict constrained minimum.}}$$

Final Answer

$$\boxed{(x_1^*, x_2^*, x_3^*) = \left(5, \frac{10}{3}, \frac{20}{3}\right)}$$

$$\boxed{\text{This is a strict constrained minimum.}}$$

1 NLPP with inequality constraints

KKT Conditions for NLPP

1. Minimization Problem

Consider the nonlinear program

$$\min_{x \in \mathbb{R}^n} \ f(x)$$

subject to

$$g_i(x) \leq 0, \quad i = 1, \ldots, m.$$

The **Karush–Kuhn–Tucker (KKT) conditions** state that a point x^* is optimal (under suitable regularity conditions) if there exist multipliers λ_i such that:

1. **Stationarity:**
$$\nabla f(x^*) = \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*).$$
2. **Primal feasibility:**
$$g_i(x^*) \leq 0, \quad i = 1, \ldots, m.$$
3. **Dual feasibility:**
$$\lambda_i \leq 0, \quad i = 1, \ldots, m.$$
4. **Complementary slackness:**
$$\lambda_i \, g_i(x^*) = 0, \quad i = 1, \ldots, m.$$

Sufficient condition for a stationary point to be optimal: The Lagrangian should be concave, i.e., the Hessian matrix of the Lagrangian should be positive definite or positive semi-definite.

2. Maximization Problem

Consider the nonlinear program

$$\max_{x \in \mathbb{R}^n} \ f(x)$$

subject to

$$g_i(x) \leq 0, \quad i = 1, \ldots, m.$$

A point x^* satisfies the KKT conditions if there exist multipliers $\lambda_i \geq 0$ and $\mu_j \in \mathbb{R}$ such that:

1. **Stationarity:**
$$\nabla f(x^*) = \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*).$$
2. **Primal feasibility:**
$$g_i(x^*) \leq 0, \quad i = 1, \ldots, m.$$

3. **Dual feasibility:**
$$\lambda_i \geq 0, \quad i = 1, \ldots, m.$$

4. **Complementary slackness:**
$$\lambda_i \, g_i(x^*) = 0, \quad i = 1, \ldots, m.$$

Sufficient condition for a stationary point to be optimal: The Lagrangian should be convex, i.e., the Hessian matrix of the Lagrangian should be negative definite or negative semi-definite.

Problem 4 *Maximize*
$$f(x_1, x_2) = 8x_1 + 10x_2 - x_1^2 - x_2^2$$
subject to the equality constraint
$$\begin{aligned} &3x_1 + 2x_2 \leq 6, \\ &x_1 \geq 0, \ x_2 \geq 0. \end{aligned}$$

Answer:

Lagrangian

Let $\lambda_i \geq 0$ be the multiplier for the constraint $g(x) = 3x_1 + 2x_2 - 6 \leq 0$, $x_1 \geq 0$ and $x_2 \geq 0$. The Lagrangian is
$$\mathcal{L} \equiv 8x_1 + 10x_2 - x_1^2 - x_2^2 = \lambda_1(3x_1 + 2x_2 - 6) - \lambda_2 x_1 - \lambda_3 x_2.$$

KKT Conditions

(1) Stationarity

$$\frac{\partial \mathcal{L}}{\partial x_1} = 3\lambda_1 - \lambda_2 \implies 8 - 2x_1 = 3\lambda_1 - \lambda_2, \tag{2}$$
$$\frac{\partial \mathcal{L}}{\partial x_2} = 2\lambda_1 - \lambda_3 \implies 10 - 2x_2 = 2\lambda_1 - \lambda_3. \tag{3}$$

(2) Primal feasibility

$$3x_1 + 2x_2 \leq 6, \qquad x_1 \geq 0, \qquad x_2 \geq 0. \tag{4}$$

(3) Dual feasibility

$$\lambda_1 \geq 0, \qquad \lambda_2 \geq 0, \qquad \lambda_3 \geq 0. \tag{5}$$

(4) Complementary slackness

$$\lambda_1(3x_1 + 2x_2 - 6) = 0, \qquad \lambda_2 x_1 = 0, \qquad \lambda_3 x_2 = 0. \tag{6}$$

Case 1: $\lambda_1 = \lambda_2 = \lambda_3 = 0$ Then from (2), (3) and (6), we have

$$x_1 = 4,\ x_2 = 5.$$

which is not satisfying the condition (4). So, this case is not possible.

Case 2: $\lambda_1 = \lambda_2 = 0,\ \lambda_3 \neq 0$ Then from last equation (6) gives

$$x_2 = 0$$

And from (2) and (3)

$$x_1 = 4,\ \lambda_3 = -10,$$

which violates the condition (5).

Case 3: $\lambda_2 = \lambda_3 = 0,\ \lambda_1 \neq 0$ From (2) and (3) gives

$$x_1 = \frac{8 - 3\lambda_1}{2},\ x_2 = \frac{10 - 2\lambda_1}{2}$$

Then from (6)

$$\lambda_1 = \frac{32}{13},\ x_1 = \frac{4}{13},\ x_2 = \frac{33}{13},$$

which satisfies all the other KKT conditions. The Hessian matrix is given by

$$H_L = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Since, $D_1 < 0, D_2 > 0$ the matrix is negative definite. Thus the optimal solution is

$$\boxed{x_1^* = \frac{4}{13}, \qquad x_2^* = \frac{33}{13}}$$

and the maximum value of objective function is

$$\boxed{f^* = \frac{277}{13}}.$$

Problem 5 *Maximize*

$$f(x_1, x_2) = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$$

subject to the equality constraint

$$\begin{aligned} &x_1 + x_2 \leq 10, \\ &x_2 \leq 8, \\ &x_1 \geq 0,\ x_2 \geq 0. \end{aligned}$$

Answer:

Lagrangian

Let $\lambda_i \geq 0$ be the multiplier for the constraint $g_1(x) = x_1+x_2-10 \leq 0$, $g_2(x) = x_2-8 \leq 0$ $g_3(x) = -x_1 \leq 0$ and $g_4(x) = -x_2 \leq 0$. The Lagrangian is

$\mathcal{L} \equiv 12x_1+21x_2+2x_1x_2-2x_1^2-2x_2^2 = \lambda_1(x_1+x_2-10)-\lambda_2\,(x_2-8)+\lambda_3(-x_1)+\lambda_4(-x_2).$

KKT Conditions

(1) Stationarity

$$\frac{\partial \mathcal{L}}{\partial x_1} = \lambda_1 - \lambda_2 - \lambda_3 \implies 12 + 2x_2 - 4x_1 = \lambda_1 - \lambda_2 - \lambda_3, \tag{7}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \lambda_1 - \lambda_2 - \lambda_4 \implies 21 - 2x_1 - 4x_2 = \lambda_1 - \lambda_2 - \lambda_4. \tag{8}$$

(2) Primal feasibility

$$x_1 + x_2 \leq 10, \qquad x_2 \leq 8, \qquad x_1 \geq 0, \qquad x_2 \geq 0. \tag{9}$$

(3) Dual feasibility

$$\lambda_1 \geq 0, \qquad \lambda_2 \geq 0, \qquad \lambda_3 \geq 0, \qquad \lambda_4 \geq 0. \tag{10}$$

(4) Complementary slackness

$$\lambda_1(x_1 + x_2 - 10) = 0, \qquad \lambda_2\,(x_2 - 8) = 0, \qquad \lambda_3(-x_1) = 0, \qquad \lambda_4(-x_2) = 0,. \tag{11}$$

Case 1: $\lambda_1 \neq 0,\ \lambda_2 \neq 0,\ \lambda_3 = 0,\ \lambda_4 = 0$ Then from (9) we have

$$x_1 = 2,\ x_2 = 8.$$

Then from (7) and (8)

$$\lambda_1 = 20,\ \lambda_2 = -27,$$

which is not satisfying the condition (8). So, this case is not possible.

Case 2: $\lambda_1 \neq 0,\ \lambda_2 = 0,\ \lambda_3 = 0,\ \lambda_4 = 0$ Then from (7), (8) and (11)

$$6x_2 - 6x_1 = 9 \qquad x_1 + x_2 = 10 \implies x_1 = \frac{69}{12}, \quad x_2 = \frac{51}{12}.$$

Using these values in (7)

$$\lambda_1 = \frac{78}{12} > 0.$$

Hence the stationary solution is $\frac{69}{12}, \frac{51}{12}, \frac{78}{12}$.
Now the Hessian matrix of L is given by

$$H_L = \begin{pmatrix} -4 & 2 \\ 2 & -4 \end{pmatrix}.$$

Since, $D_1 < 0, D_2 > 0$ the matrix is negative definite. Thus the optimal solution is

$$\boxed{x_1^* = \frac{69}{12}, \qquad x_2^* = \frac{51}{12}}$$

and the maximum value of objective function is

$$\boxed{f^* = \frac{947}{8}}.$$

Problem 6 *Minimize*

$$f(x_1, x_2) = x_1^2 + x_2^2$$

subject to the equality constraint

$$\begin{aligned} &2x_1 + x_2 \leq 5, \\ &x_1 \geq 1, \\ &x_2 \geq 0. \end{aligned}$$

Answer:

Lagrangian

Since, the given problem is minimization type, we consider $\lambda_i \leq 0$ be the multiplier for the constraint $g_1(x) = 2x_1 + x_2 - 5 \leq 0$, $g_2(x) = -x_1 + 1 \leq 0$ and $g_3(x) = -x_2 \leq 0$. The Lagrangian is

$$\mathcal{L} \equiv x_1^2 + x_2^2 = \lambda_1(2x_1 + x_2 - 5) - \lambda_2(-x_1 + 1) + \lambda_3(-x_2).$$

KKT Conditions

(1) Stationarity

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2\lambda_1 + \lambda_2 \implies 2x_1 = 2\lambda_1 + \lambda_2, \tag{12}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \lambda_1 - \lambda_3 \implies 2x_2 = \lambda_1 - \lambda_3. \tag{13}$$

(2) Primal feasibility

$$2x_1 + x_2 \leq 5, \qquad x_1 \geq 1, \qquad x_2 \geq 0. \tag{14}$$

(3) Dual feasibility

$$\lambda_1 \leq 0, \qquad \lambda_2 \leq 0, \qquad \lambda_3 \leq 0. \tag{15}$$

(4) Complementary slackness

$$\lambda_1(2x_1 + x_2 - 5) = 0, \qquad \lambda_2(-x_1 + 1) = 0, \qquad \lambda_3(-x_2) = 0, . \tag{16}$$

Case 1: $\lambda_1 \neq 0,\ \lambda_2 \neq 0,\ \lambda_3 = 0$ Then from (9) we have

$$x_1 = 1,\ x_2 = 3.$$

Then from (12)

$$\lambda_1 = 6,$$

which is not satisfying the condition (15). So, this case is not possible.

Case 2: $\lambda_1 \neq 0,\ \lambda_2 = 0,\ \lambda_3 = 0$ Then from (12) and (13) we have

$$x_1 = \lambda_1,\ x_2 = \frac{\lambda_1}{2}.$$

Using these values in (16)

$$\lambda_1 = 2,$$

which is not satisfying the condition (15). So, this case is not possible.

Case 3: $\lambda_1 = 0,\ \lambda_2 \neq 0,\ \lambda_3 = 0$ Which gives $x_1 = 1,\ x_2 = 0,\ \lambda_2 = -2$. Now the Hessian matrix of L is given by

$$H_L = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Since, $D_1 > 0, D_2 > 0$ the matrix is positive definite. Thus the optimal solution is

$$\boxed{x_1^* = 1, \qquad x_2^* = 0}$$

and the maximum value of objective function is

$$\boxed{f^* = 1}.$$